

Exercise 4.

a)

$$g(0) = 0$$

$$g(1) = 5$$

$$g(2) = 3$$

$$g(n+3) = 2g(n+2) + g(n+1) - 2g(n)$$

$$g(n) = \Theta(a^n)$$

$$g(n) = a^n$$

$$a^{n+3} = 2a^{n+2} + a^{n+1} - 2a^n, \text{ knowing that } a \neq 0$$

$$\text{we get } a^3 = 2a^2 + a - 2$$

$$a^3 - 2a^2 - a + 2 = 0$$

$$a_1 = -1 \vee a_2 = 1 \vee a_3 = 2$$

Try to find c_1, c_2, c_3 such that

$$g(n) = c_1 a_1^n + c_2 a_2^n + c_3 a_3^n$$

$$g(n) = c_1 (-1)^n + c_2 \cdot 1^n + c_3 \cdot 2^n$$

$$g(n) = (-1)^n c_1 + c_2 + 2^n c_3$$

$$\begin{cases} g(0) = 0 \\ g(1) = 5 \\ g(2) = 3 \end{cases}$$

$$\begin{cases} (-1)^0 c_1 + c_2 + 2^0 c_3 = 0 \\ (-1)^1 c_1 + c_2 + 2^1 c_3 = 5 \\ (-1)^2 c_1 + c_2 + 2^2 c_3 = 3 \end{cases}$$

$$\begin{cases} c_1 + c_2 + c_3 = 0 \\ -c_1 + c_2 + 2c_3 = 5 \\ c_1 + c_2 + 4c_3 = 3 \end{cases}$$

$$\begin{cases} c_1 = -c_2 - c_3 \\ c_2 + c_3 + c_2 + 2c_3 = 5 \\ -c_2 - c_3 + c_2 + 4c_3 = 3 \end{cases}$$

$$\begin{cases} c_1 = -c_2 - c_3 \\ 2c_2 + 3c_3 = 5 \\ 3c_3 = 3 \end{cases}$$

$$\begin{cases} C_1 = -C_2 - 1 \\ 2C_2 + 3 = 5 \\ C_3 = 1 \end{cases}$$

$$\begin{cases} C_1 = -C_2 - 1 \\ 2C_2 = 2 \\ C_3 = 1 \end{cases}$$

$$\begin{cases} C_1 = -1 - 1 \\ C_2 = 1 \\ C_3 = 1 \end{cases}$$

$$\begin{cases} C_1 = -2 \\ C_2 = 1 \\ C_3 = 1 \end{cases}$$

$$\text{so } \boxed{g(n) = -2 \cdot (-1)^n + 1^n + 2^n}$$

6) $g(n) \in \Theta(2^n)$ if it's also in $O(2^n)$ and $\Omega(2^n)$. We can say that $-2 \cdot (-1)^n + 1^n$ can be omitted for complexity with a large enough input.

Proof: $O(2^n)$: $2^n \leq c \cdot 2^n$ for $c \geq 1$
 $1 \geq c \geq 0$

Proof: $\Omega(2^n)$: $2^n \geq c \cdot 2^n$ for $c \leq 1$

Exercise 5.

a) $T(n) = 4T(\lfloor \frac{n}{6} \rfloor) + T(\lfloor \frac{n}{3} \rfloor) + dn$

Proof of $T(n) = O(n \log n)$

We need to find N and C such that $T(n) \leq Cn \log n$ for all $n > N$. We will determine these along the way of the proof. For $n \geq 1$ we have:

$$\begin{aligned}
 T(n) &= 4T(\lfloor \frac{n}{6} \rfloor) + T(\lfloor \frac{n}{3} \rfloor) + dn \\
 &\stackrel{IH}{\leq} 4 \cdot C(\lfloor \frac{n}{6} \rfloor) \log(\lfloor \frac{n}{6} \rfloor) + C(\lfloor \frac{n}{3} \rfloor) \log(\lfloor \frac{n}{3} \rfloor) + dn \\
 &\leq 4C \cdot (\frac{n}{6}) \log(\frac{n}{6}) + C \cdot (\frac{n}{3}) \log(\frac{n}{3}) + dn
 \end{aligned}$$

$$\begin{aligned}
 &= 4C \cdot (\frac{n}{6}) \log n - 4C \cdot (\frac{n}{6}) \log 6 + C \cdot (\frac{n}{3}) \log n - C \cdot (\frac{n}{3}) \log 3 + dn \\
 &= 2C \cdot (\frac{n}{3}) \log n - 2C \cdot (\frac{n}{3}) \log 6 + C \cdot (\frac{n}{3}) \log n - C \cdot (\frac{n}{3}) \log 3 + dn \\
 &= C \cdot (\frac{n}{3}) (2 \log n - 2 \log 6 + \log n - \log 3) + dn \\
 &= \frac{1}{3} Cn (3 \log n - 2 \log 2 - 2 \log 3 - \log 3) + dn \\
 &= \frac{1}{3} Cn (3 \log n - 2 \log 2 - 3 \log 3) + dn \\
 &= Cn \log n + \frac{1}{3} Cn (-2 \log 2 - 3 \log 3) + dn \\
 &= Cn \log n - \frac{2}{3} Cn \log 2 - Cn \log 3 + dn \\
 (*) &\leq Cn \log n
 \end{aligned}$$

(*) we have to make sure that

$$\frac{2}{3} Cn \log 2 + Cn \log 3 \geq dn$$

We have

$$\begin{aligned}
 \frac{2}{3} Cn \log 2 + Cn \log 3 &\geq \frac{2}{3} Cn \log 2 + Cn \log 2 \\
 &= \frac{5}{3} Cn \log 2
 \end{aligned}$$

so we can choose C such that

$$\frac{5}{3} C n \log 2 \geq d n$$

$$\frac{5}{3} C \log 2 \geq d$$

$$5 C \log 2 \geq 3 d$$

$$C \geq \frac{3d}{5 \log 2}$$

and C large enough so that also for the base cases for n we have

$T(n) \leq C n \log n$. There is no restriction on N , so we can choose $N := 1$

Exercise 7.

$$a) T(n) = 3T(n-1) + 4T(n-2) + \Theta(n^3)$$

Let N_0, C_0 be such that for all $n \geq N_0$ $T(n) \leq 3T(n-1) + 4T(n-2) + C_0 n^3$. Take $N := N_0$ and C such that C is sufficiently large. A direct proof by induction doesn't work, so we let $d \geq \frac{C_0}{4}$ and prove $T(n) \leq C 4^n - d n^3$

$$T(n) \leq 3T(n-1) + 4T(n-2) + C_0 n^3$$

$$\stackrel{IH}{\leq} 3 \cdot (4^{n-1}C - d(n-1)^3) + 4 \cdot (4^{n-2}C - d(n-2)^3) + C_0 n^3$$

$$= 3 \cdot 4^{n-1}C - 3d(n-1)^3 + 4^{n-1}C - 4d(n-2)^3 + C_0 n^3$$

$$= 4^{n-1}C \cdot (3+1) - 3d(n-1)^3 - 4d(n-2)^3 + C_0 n^3$$

$$= 4^n C - 3d(n^3 - 3n^2 + 3n - 1) - 4d(n^3 - 6n^2 + 12n - 8) + C_0 n^3$$

$$= 4^n C - 3dn^3 + 9dn^2 - 9dn + 3d - 4dn^3 + 24dn^2 - 48dn + 32d + C_0 n^3$$

$$= 4^n C - 7dn^3 + 33dn^2 - 57dn + 35d + C_0 n^3$$

$$= 4^n C + n^3(C_0 - 7d) + 33dn^2 - 57dn + 35d$$

if $C_0 \leq 4d$ then

$$\leq 4^n C + n^3(4d - 7d) + 33dn^2 - 57dn + 35d$$

$$= 4^n C - 3dn^3 + 33dn^2 - 57dn + 35d$$

$$= 4^n C - d(3n^3 - 33n^2 + 57n - 35)$$

$$\leq 4^n C - dn^3$$

for all C , for sufficiently large n because

$$3n^3 - 33n^2 + 57n - 35 \geq n^3 \text{ eventually}$$

to be precise if $n \geq 15$

b) Let C_0, N_0 be such that for all $n \geq N_0$ $T(n) = 3T(n-1) + 4T(n-2) + C_0 n^3$.
We want to prove $T(n) \geq C \cdot 4^n$ for some C and n sufficiently large.

$$\begin{aligned} T(n) &\geq 3T(n-1) + 4T(n-2) + C_0 n^3 \\ &\stackrel{IH}{\geq} 3 \cdot (4^{n-1} \cdot C) + 4 \cdot (4^{n-2} \cdot C) + C_0 n^3 \\ &= \cancel{3 \cdot 4^n \cdot C} \end{aligned}$$

$$\begin{aligned} &= 3 \cdot 4^{n-1} \cdot C + 4^{n-1} \cdot C + C_0 n^3 \\ &= 3 \cdot 4^n \cdot 4^{-1} \cdot C + 4^n \cdot 4^{-1} \cdot C + C_0 n^3 \\ &= 4^n \cdot C \left(3 \cdot \frac{1}{4} + \frac{1}{4} \right) + C_0 n^3 \\ &= 4^n \cdot C \cdot 1 + C_0 n^3 \\ &= 4^n \cdot C + C_0 n^3 \\ &\geq 4^n C \end{aligned}$$

for all C . We choose C small enough so that also for the base cases for n we have $T(n) \geq 4^n C$.